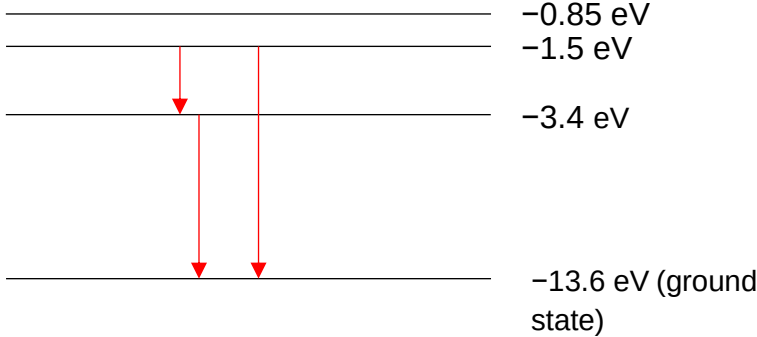
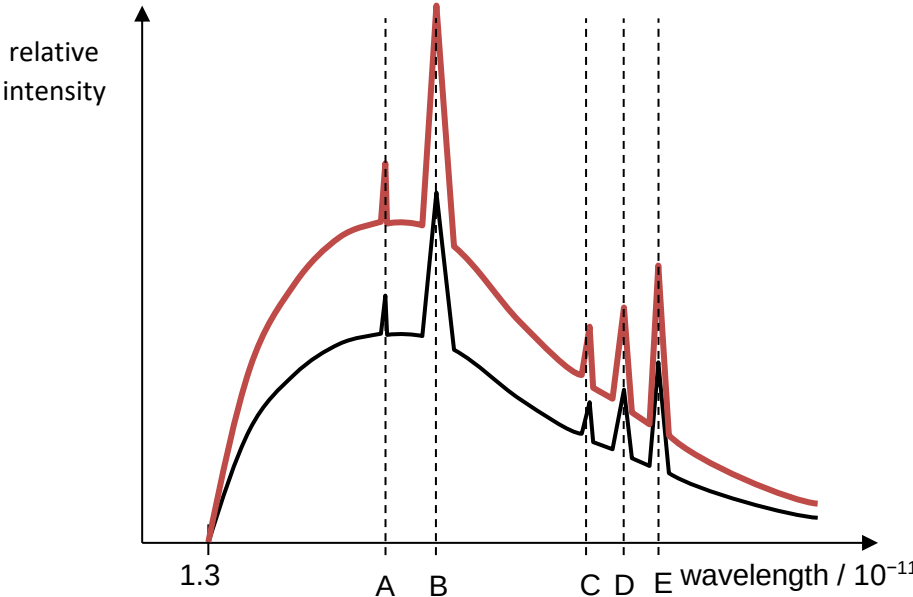


6	(a)	(i)	Emission line spectrum consists of <u>discrete bright coloured lines</u> on a <u>dark background</u>. [B1] Absorption spectrum consists of <u>dark lines</u> against a <u>continuous spectrum of white light</u>. [B1]
		(ii)	Summary of important points: 1. <u>Photons emitted</u> due to <u>de-excitation</u> [B1] 2. <u>Frequency/wavelength</u> of photon corresponds to difference between <u>discrete energy levels</u>. [B1] 3. Prism <u>separates the photons</u> according to wavelength/frequency [B1] <u>Sample response</u> The excited gas atoms are unstable. When the gas electrons <u>de-excite</u> a lower energy level, <u>the energy is emitted as photons (electromagnetic radiation)</u> with a specific frequency. [B1] The frequency f of the emission line is dependent on the difference between the high and low energy levels, $E = hf$. Due to the <u>discrete energy levels</u>, only certain high-to-low energy level transitions are possible within the atom, therefore <u>only certain frequencies</u> of photons are emitted. [B1] The prism then <u>separates the photons according to wavelength/frequencies</u>, resulting in the line spectrum [B1]
		(iii)	 [B1]

	(iv)	 increasing frequency $\longrightarrow$ [B1] the energy of the photon emitted by the electron de-exciting from level 2 to ground state, and level 1 to ground state, are more similar in magnitude. Hence, their frequencies are closer, leading to their lines being near each other. As such, the de-excitation from level 2 to level 1 is further away. Note: ECF due to drawing 6 de-excitations from level 3 in (iii)  increasing frequency $\longrightarrow$
(b)	(i)	$E = \frac{hc}{\lambda}$ $= \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{(1.3 \times 10^{-11})} \quad [\text{B1}]$ $= 1.53 \times 10^{-14}$ $\approx 1.5 \times 10^{-14} \text{ J}$
	(ii)	Finding velocity of the electron $\frac{1}{2}mv^2 = E_{\text{photon max}}$ $v = \sqrt{\frac{2E_{\text{photon max}}}{m}}$ $= \sqrt{\frac{2(1.5 \times 10^{-14})}{9.11 \times 10^{-31}}}$ $= 1.8146 \times 10^8 \text{ m s}^{-1} \quad [\text{C1}]$ finding momentum $p_e = mv$ $= (9.11 \times 10^{-31})(1.8146 \times 10^8)$ $= 1.65 \times 10^{-22} \text{ kg m s}^{-1} \quad [\text{A1}]$

		(iii) finding de Broglie wavelength $\lambda = \frac{h}{p_e}$ $= \frac{(6.63 \times 10^{-34})}{(1.6531 \times 10^{-22})} \quad [C1]$ $= 4.0106 \times 10^{-12}$ $= 4.01 \times 10^{-12} \text{ m} \quad [A1]$
		(iv) highest probability – B [B1]
		(v)  relative intensity 1.3 A B C D E wavelength / 10^{-11} same minimum wavelength, [B1] all points above the previous graph with characteristic peaks at same locations [B1]

7	(a)	(i)	The central maximum is white in colour while the first order maxima on
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